

## 4.1 Answer each of the following:

- a. Suppose that a simple regression has quantities  $N = 20$ ,  $\sum y_i^2 = 7825.94$ ,  $\bar{y} = 19.21$ , and  $SSR = 375.47$ , find  $R^2$ .
- b. Suppose that a simple regression has quantities  $R^2 = 0.7911$ ,  $SST = 725.94$ , and  $N = 20$ , find  $\hat{\sigma}^2$ .
- c. Suppose that a simple regression has quantities  $\sum (y_i - \bar{y})^2 = 631.63$  and  $\sum \hat{e}_i^2 = 182.85$ , find  $R^2$ .

$$\begin{aligned} \text{a. } SST &= \sum y_i^2 - N \bar{y}^2 \\ &= 7825.94 - 20 \times (19.21)^2 = 445.458 \\ R^2 &= \frac{SSR}{SST} = \frac{375.47}{445.458} = 0.8429 \end{aligned}$$

$$\begin{aligned} \text{b. } R^2 &= \frac{SSR}{SST} = \frac{SSR}{725.94} = 0.7911, \quad SSR = 0.7911 \times 725.94 = 574.2911 \\ SST &= SSR + SSE \Rightarrow 725.94 = 574.2911 + SSE \\ SSE &= 725.94 - 574.2911 = 151.6489 = \sum \hat{e}_i^2 \\ \hat{\sigma}^2 &= \frac{\sum \hat{e}_i^2}{N-2} = \frac{151.6489}{20-2} = 8.4249 \end{aligned}$$

$$\begin{aligned} \text{c. } \sum (y_i - \bar{y})^2 &= SST = 631.63 \\ \sum \hat{e}_i^2 &= SSE = 182.85 \\ R^2 &= 1 - \frac{SSE}{SST} = 1 - \frac{182.85}{631.63} = 0.7105 \end{aligned}$$

4.3 We have five observations on  $x$  and  $y$ . They are  $x_i = 3, 2, 1, -1, 0$  with corresponding  $y$  values  $y_i = 4, 2, 3, 1, 0$ . The fitted least squares line is  $\hat{y}_i = 1.2 + 0.8x_i$ , the sum of squared least squares residuals is  $\sum_{i=1}^5 \hat{e}_i^2 = 3.6$ ,  $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$ , and  $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$ . Carry out this exercise with a hand calculator. Compute

- the predicted value of  $y$  for  $x_0 = 4$ .
- the  $se(f)$  corresponding to part (a).
- a 95% prediction interval for  $y$  given  $x_0 = 4$ .
- a 99% prediction interval for  $y$  given  $x_0 = 4$ .
- a 95% prediction interval for  $y$  given  $x = \bar{x}$ . Compare the width of this interval to the one computed in part (c).

a.  $\hat{y}_0 = 1.2 + 0.8 \times 4 = 4.4$

b.  $\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{5-2} = 1.2$   
 $\bar{x} = \frac{\sum x_i}{N} = \frac{3+2+1+(-1)+0}{5} = 1$   
 $se(f) = \sqrt{\widehat{var}(f)} = \sqrt{\hat{\sigma}^2 \left[ 1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]} = \sqrt{1.2 \left( 1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right)} = 1.5875$

c.  $\alpha = 1 - 95\% = 0.05$ ,  $df = 5 - 2 = 3$

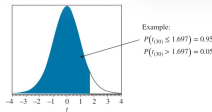
由查表可知  $t_{\alpha/2, df} = 3.182$

$\hat{y}_0 = 4.4$ ,  $se(f) = 1.5875$

$\Rightarrow 95\% \text{ prediction interval: } \hat{y}_0 \pm t_{n-2, \alpha/2} se(f)$

$\Rightarrow [4.4 - 3.182 \times 1.5875, 4.4 + 3.182 \times 1.5875]$

$= [-0.6514, 4.4514]$



| df | $t_{0.50, df}$ | $t_{0.25, df}$ | $t_{0.10, df}$ | $t_{0.05, df}$ | $t_{0.025, df}$ |
|----|----------------|----------------|----------------|----------------|-----------------|
| 1  | 3.078          | 6.314          | 12.706         | 31.821         | 63.657          |
| 2  | 1.886          | 2.920          | 4.303          | 6.965          | 9.925           |
| 3  | 1.638          | 2.353          | 3.182          | 4.541          | 5.841           |
| 4  | 1.533          | 2.132          | 2.776          | 3.747          | 4.604           |
| 5  | 1.476          | 2.015          | 2.571          | 3.365          | 4.032           |

d.  $\alpha = 1 - 99\% = 0.01$ ,  $df = 5 - 2 = 3$

由查表可知  $t_{\alpha/2, df} = 5.841$

$\hat{y}_0 = 4.4$ ,  $se(f) = 1.5875$

$\Rightarrow 99\% \text{ prediction interval: } \hat{y}_0 \pm t_{n-2, \alpha/2} se(f)$

$\Rightarrow [4.4 - 5.841 \times 1.5875, 4.4 + 5.841 \times 1.5875]$

$= [-4.8726, 13.6726]$

e. 由 b 小题可知  $\bar{x} = 1$ ,  $\hat{\sigma}^2 = 1.2$

$\hat{y}_0 = 1.2 + 0.8 \times 1 = 2$

$se(f) = \sqrt{\widehat{var}(f)} = \sqrt{\hat{\sigma}^2 \left[ 1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]} = \sqrt{1.2 \left( 1 + \frac{1}{5} + \frac{(1-1)^2}{10} \right)} = 1.2$

$\alpha = 1 - 95\% = 0.05$ ,  $df = 5 - 2 = 3$ , 由查表可知  $t_{\alpha/2, df} = 3.182$

$\hat{y}_0 = 2$ ,  $se(f) = 1.5875$

$\Rightarrow 95\% \text{ prediction interval: } \hat{y}_0 \pm t_{n-2, \alpha/2} se(f)$

$\Rightarrow [2 - 3.182 \times 1.2, 2 + 3.182 \times 1.2]$

$= [-1.8184, 5.8184]$

width of interval in part c:  $4.4514 - (-0.6514) = 5.1028$

width of interval in part e:  $5.8184 - (-1.8184) = 7.6368$

$\Rightarrow$  prediction interval in  $x = \bar{x}$  時會小於在  $x = 4$  時的。

$\because$  當把  $x$  設定於不在樣本範圍 ( $x_i = 3, 2, 1, -1, 0$ ) 內, 即  $x_0 = 4$  時, 模型在此值時做預測的不確定性較高, 導致 prediction interval 較大。

4.25 The file *collegetown* contains data on 500 houses sold in Baton Rouge, LA during 2009–2013. Variable descriptions are in the file *collegetown.def*.

- Estimate the log-linear model  $\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$ . Interpret the estimated model parameters. Calculate the slope and elasticity at the sample means, if necessary.
- Estimate the log-log model  $\ln(PRICE) = \alpha_1 + \alpha_2 \ln(SQFT) + e$ . Interpret the estimated parameters. Calculate the slope and elasticity at the sample means, if necessary.
- Compare the  $R^2$  value from the linear model  $PRICE = \delta_1 + \delta_2 SQFT + e$  to the “generalized”  $R^2$  measure for the models in (b) and (c).
- Construct histograms of the least squares residuals from each of the models in (a)–(c) and obtain the Jarque–Bera statistics. Based on your observations, do you consider the distributions of the residuals to be compatible with an assumption of normality?
- For each of the models in (a)–(c), plot the least squares residuals against *SQFT*. Do you observe any patterns?
- For each model in (a)–(c), predict the value of a house with 2700 square feet.
- For each model in (a)–(c), construct a 95% prediction interval for the value of a house with 2700 square feet.
- Based on your work in this problem, discuss the choice of functional form. Which functional form would you use? Explain.

a.

```
> library(POESrdata)
> library(ggplot2)
> data("collegetown")
> ?data("collegetown")
>
> # (a)
> collegetown$ln_price <- log(collegetown$price)
> model_loglinear <- lm(ln_price ~ sqft, data = collegetown)
> summary(model_loglinear)

Call:
lm(formula = ln_price ~ sqft, data = collegetown)

Residuals:
    Min       1Q   Median       3Q      Max
-1.32528 -0.19199  0.00122  0.21678  0.86609

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.393866   0.043288  101.50  <2e-16 ***
sqft         0.036044   0.001486   24.26  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.34 on 498 degrees of freedom  
Multiple R-squared: 0.5417, Adjusted R-squared: 0.5408  
F-statistic: 588.7 on 1 and 498 DF, p-value: < 2.2e-16

log-linear model  $\Rightarrow \ln(PRICE) = 4.393866 + 0.036044 SQFT$

A1: When  $SQFT = 0$ , the value of  $\ln(PRICE)$  is 4.393866

A2: When  $SQFT$  increase by 1 unit,  $PRICE$  will increase by 0.036044%.

```
> beta1 <- coef(model_loglinear)[1]
> beta2 <- coef(model_loglinear)[2]
>
> # sample mean
> mean_sqft <- mean(collegetown$sqft)
> cat("Sample mean for SQFT:", mean_sqft, "\n")
Sample mean for SQFT: 27.28212
> mean_price <- mean(collegetown$price)
> cat("Sample mean for PRICE:", mean_price, "\n")
Sample mean for PRICE: 250.2369
>
> # slope at sample mean
> # d(price)/d(sqft) = beta2 * price
> slope_at_mean_loglinear <- beta2 * mean_price
> cat("Slope at sample means:", slope_at_mean_loglinear, "\n")
Slope at sample means: 9.019661
>
> # elasticity at sample mean
> elasticity_at_mean_loglinear <- beta2 * mean_sqft
> cat("Elasticity at sample means:", elasticity_at_mean_loglinear, "\n")
Elasticity at sample means: 0.9833701
```

b.

```
> # (b)
> collegetown$ln_sqft <- log(collegetown$sqft)
> model_loglog <- lm(ln_price ~ ln_sqft, data = collegetown)
> summary(model_loglog)

Call:
lm(formula = ln_price ~ ln_sqft, data = collegetown)

Residuals:
    Min       1Q   Median       3Q      Max
-1.36864 -0.20849  0.00487  0.23204  1.21099

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.04965   0.15797   12.97  <2e-16 ***
ln_sqft      1.02483   0.04839   21.18  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3643 on 498 degrees of freedom
Multiple R-squared: 0.4738, Adjusted R-squared: 0.4728
F-statistic: 448.5 on 1 and 498 DF, p-value: < 2.2e-16
```

log-log model  $\Rightarrow \ln(PRICE) = 2.04965 + 1.02483 \ln(SQFT)$

A1: When  $\ln(SQFT) = 0$ , the value of  $\ln(PRICE)$  is 2.04965

A2: When  $SQFT$  increase by 1%,  $PRICE$  will increase by 1.02483%.

```
> alpha1 <- coef(model_loglog)[1]
> alpha2 <- coef(model_loglog)[2]
>
> # sample mean
> mean_sqft <- mean(collegetown$sqft)
> cat("Sample mean for SQFT:", mean_sqft, "\n")
Sample mean for SQFT: 27.28212
> mean_price <- mean(collegetown$price)
> cat("Sample mean for PRICE:", mean_price, "\n")
Sample mean for PRICE: 250.2369
>
> # slope at sample means
> # d(price)/d(sqft) = alpha2 * (price / sqft)
> slope_at_mean_loglog <- alpha2 * (mean_price / mean_sqft)
> cat("Slope at mean:", slope_at_mean_loglog, "\n")
Slope at mean: 9.399923
>
> # elasticity at sample means
> elasticity_loglog <- alpha2
> cat("Elasticity (alpha2):", elasticity_loglog, "\n")
Elasticity (alpha2): 1.024828
```

C.

```
> #log-linear model
> log.linear = lm(log(price)~sqft, data = collettown)
> log.linear

Call:
lm(formula = log(price) ~ sqft, data = collettown)

Coefficients:
(Intercept)      sqft
      4.39387      0.03604
```

```
> sum_log.linear = summary(log.linear)
> #R^2
> sum_log.linear$squared
[1] 0.541729
> #general_R^2
> y_hat = exp(coef(log.linear)[1]+coef(log.linear)[2]*collettown$sqft)
> y = collettown$price
> general_R = cor(y,y_hat)^2
> general_R
[1] 0.6621612
```

```
> #log-log model
> log.log = lm(log(price)~log(sqft), data = collettown)
> log.log

Call:
lm(formula = log(price) ~ log(sqft), data = collettown)

Coefficients:
(Intercept)    log(sqft)
      2.050      1.005
```

```
> sum_log.log = summary(log.log)
> #R^2
> sum_log.log$squared
[1] 0.4738445
> #general_R^2
> y_hat = exp(coef(log.log)[1]+coef(log.log)[2]*log(collettown$sqft))
> y = collettown$price
> general_R = cor(y,y_hat)^2
> general_R
[1] 0.645584
```

```
> #linear model
> linear = lm(price~sqft, data = collettown)
> linear

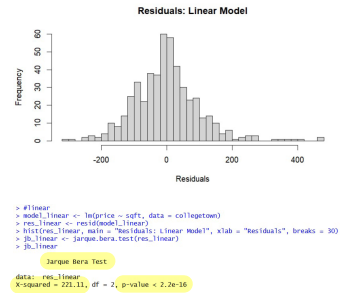
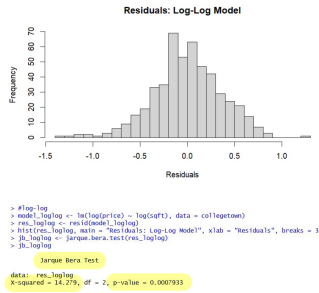
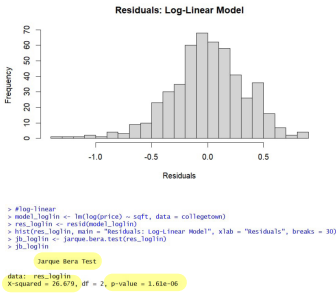
Call:
lm(formula = price ~ sqft, data = collettown)

Coefficients:
(Intercept)      sqft
     -115.4      13.4
```

```
> sum_linear = summary(linear)
> #R^2
> sum_linear$squared
[1] 0.643167
> #general_R^2
> y_hat = coef(linear)[1]+coef(linear)[2]*collettown$sqft
> y = collettown$price
> general_R = cor(y,y_hat)^2
> general_R
[1] 0.643167
```

| Model      | $R^2$     | generalized $R^2$ |
|------------|-----------|-------------------|
| log-log    | 0.541729  | 0.6621612         |
| log-linear | 0.4738445 | 0.645584          |
| linear     | 0.643167  | 0.643167          |

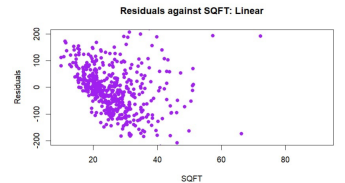
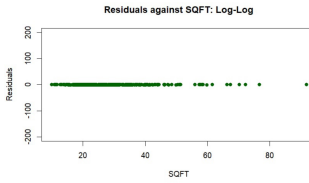
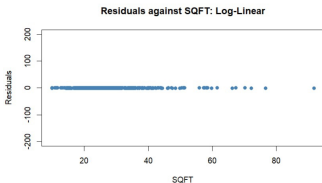
d.



Jarque-Bera test 是用来检测资料是否服从常態分布  $H_0$ 。资料是常態分布的。  $H_1$  资料不是常態分布的。

上述资料结果中的 p-value 都  $< 5\%$   $\rightarrow$  拒绝  $H_0 \rightarrow$  资料不是常態分布的。

e.



log-linear 和 log-log 模型的 residual 波动比 linear model 的小,  
且在给定 SQFT 时, 这两种模型的期望 residual value 为 0, 但 linear model 就无此特徵。

f.

```

> sqft_f=2700/100
> model_loglinear <- lm(ln_price ~ sqft, data = collegietown)
> b1=coef(model_loglinear)[1]]
> b2=coef(model_loglinear)[2]]
> y1=exp(b1+b2*sqft_f)
> cat("Predicted value for log-linear model:", y1*1000, "\n")
Predicted value for log-linear model: 214233.6
>
> model_loglog <- lm(ln_price ~ ln_sqft, data = collegietown)
> a1=coef(model_loglog)[1]]
> a2=coef(model_loglog)[2]]
> y2=exp(a1+a2*ln(sqft_f))
> cat("Predicted value for log-log model:", y2*1000, "\n")
Predicted value for log-log model: 227538.6
>
> model_linear <- lm(price ~ sqft, data = collegietown)
> d1=coef(model_linear)[1]]
> d2=coef(model_linear)[2]]
> y3=d1+d2*sqft_f
> cat("Predicted value for linear model:", y3*1000, "\n")
Predicted value for linear model: 246455.7

```

g.

```

> sqft = collegietown$sqft
> price = collegietown$price
> ln_price = log(price)
> ln_sqft = log(sqft)
> mean_sqft = mean(sqft)
> model_loglinear <- lm(ln_price ~ sqft, data = collegietown)
> b1=coef(model_loglinear)[1]]
> b2=coef(model_loglinear)[2]]
> model_loglog <- lm(ln_price ~ ln_sqft, data = collegietown)
> a1=coef(model_loglog)[1]]
> a2=coef(model_loglog)[2]]
> model_linear <- lm(price ~ sqft, data = collegietown)
> d1=coef(model_linear)[1]]
> d2=coef(model_linear)[2]]
> price_1_hat = b1 + 27*82
> price_2_hat = a1 + log(27)*a2
> price_3_hat = d1 + 27*d2
>
> tvalue = qt(0.975,498)
> vara_1_2 = vcov(model_loglinear)[2,2]
> se1 = summary(model_loglinear)
> sigma_hat_1 = sn1$sigma^2
> varf_1 = sigma_hat_1 + sigma_hat_1/500 + (27 - mean_sqft)^2 * vara_1_2
> sef_1 = sqrt(varf_1)
> lowb_1 = exp(as.numeric(price_1_hat) - tvalue*sef_1)
> upb_1 = exp(as.numeric(price_1_hat) + tvalue * sef_1)
> cat("Prediction interval for log-linear model:", lowb_1,"",upb_1, "\n")
Prediction interval for log-linear model: 109.7685 , 418.1167
> tvalue = qt(0.975,498)
> vara_2_2 = vcov(model_loglog)[2,2]
> se2 = summary(model_loglog)
> sigma_hat_2 = sn2$sigma^2
> varf_2 = sigma_hat_2 + sigma_hat_2/500 + (27 - mean_sqft)^2 * vara_2_2
> sef_2 = sqrt(varf_2)
> lowb_2 = exp(as.numeric(price_2_hat) - tvalue * sef_2)
> upb_2 = exp(as.numeric(price_2_hat) + tvalue * sef_2)
> cat("Prediction interval for log-log model:", lowb_2,"",upb_2, "\n")
Prediction interval for log-log model: 111.0866 , 466.0674
> tvalue = qt(0.975,498)
> vara_3_2 = vcov(model_linear)[2,2]
> se3 = summary(model_linear)
> sigma_hat_3 = sn3$sigma^2
> varf_3 = sigma_hat_3 + sigma_hat_3/500 + (27 - mean_sqft)^2 * vara_3_2
> sef_3 = sqrt(varf_3)
> lowb_3 = as.numeric(price_3_hat) - tvalue * sef_3
> upb_3 = as.numeric(price_3_hat) + tvalue * sef_3
> cat("Prediction interval for linear model:", lowb_3,"",upb_3, "\n")
Prediction interval for linear model: 44.2727 , 448.6341

```

h.

```

> model_loglinear <- lm(ln_price ~ sqft, data = collegietown)
> pred_loglinear <- predict(model_loglinear)
> #prediction result
> pred_loglinear <- c(pred_loglinear)
>
> model_loglog <- lm(ln_price ~ ln_sqft, data = collegietown)
> pred_loglog <- exp(predict(model_loglog))
> pred_loglog <- c(pred_loglog)
>
> model_linear <- lm(price ~ sqft, data = collegietown)
> pred_linear <- predict(model_linear)
> pred_linear <- c(pred_linear)
>
> #actual value
> actual <- c(collegietown$price)
>
> #model RMSE
> calculate_rmse <- function(pred, actual) {
+   rmse <- sqrt(mean((pred - actual)^2))
+   return(rmse)
+ }
>
> #model RMSE
> rmse_loglinear <- calculate_rmse(pred_loglinear, actual)
> rmse_loglog <- calculate_rmse(pred_loglog, actual)
> rmse_linear <- calculate_rmse(pred_linear, actual)
> cat(paste("Log-linear model's RMSE:", rmse_loglinear, "\n"))
Log-linear model's RMSE: 298.664589187882
> cat(paste("Log-log model's RMSE:", rmse_loglog, "\n"))
Log-log model's RMSE: 114.851531313631
> cat(paste("Linear model's RMSE:", rmse_linear, "\n"))
Linear model's RMSE: 102.594804421168
> #model MSE
> calculate_mse <- function(pred, actual) {
+   mse <- mean((pred - actual)^2)
+   return(mse)
+ }
> mse_loglinear <- calculate_mse(pred_loglinear, actual)
> mse_loglog <- calculate_mse(pred_loglog, actual)
> mse_linear <- calculate_mse(pred_linear, actual)
> cat(paste("Log-linear model's MSE:", mse_loglinear, "\n"))
Log-linear model's MSE: 89200.5368347664
> cat(paste("Log-log model's MSE:", mse_loglog, "\n"))
Log-log model's MSE: 13190.8742450859
> cat(paste("Linear model's MSE:", mse_linear, "\n"))
Linear model's MSE: 10525.6938942177

```

| Model      | $R^2$                  | generalized $R^2$      | Jarque-Bera<br>p-value | residual<br>分布  | MSE                    | RMSE                 |
|------------|------------------------|------------------------|------------------------|-----------------|------------------------|----------------------|
| log-log    | 0.5419259              | 0.6621612 <sup>√</sup> | 1.61e <sup>-6</sup>    | 收敛 <sup>√</sup> | 89200.539              | 298.664              |
| log-linear | 0.4438445              | 0.6445844              | 0.0007933 <sup>√</sup> | 收敛 <sup>√</sup> | 13190.894              | 114.852              |
| linear     | 0.6413167 <sup>√</sup> | 0.6413167              | <2.2e <sup>-6</sup>    | 发散              | 10525.694 <sup>√</sup> | 105.595 <sup>√</sup> |

⇒ linear model 的 predict 能力最佳, log-linear 和 log-log 的残差分布较理想.

我会选择 linear model 来做更好的 prediction.